

EXAMSPHERE

Controller of Examination Portal

Simple & Detailed Prototype Document

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From Registration to Results — One Secure Examination Ecosystem.

1. Problem Statement

Design and develop a comprehensive Controller of Examination portal that digitizes manual processes, provides role-based access for administrators, staff and users, automates workflows, generates analytical reports, sends real-time notifications and integrates securely with institutional systems.

The aim is to improve operational efficiency, transparency, collaboration and user experience throughout the examination process.

2. Problems in the Current System

Problem	What happens
Manual work	Registration, seating, attendance, marks and reports take more time.
Scattered data	Student, course, exam and result information may be stored in different places.
Communication delay	Students and staff may not receive timetable, hall or result updates quickly.
Human errors	Manual calculations and data entry can cause mistakes.
Difficult monitoring	COE cannot easily see the complete examination status in one place.
Security issues	Question papers, marks and certificates need controlled access.

3. Proposed Solution — ExamSphere

ExamSphere is a single digital platform that manages the complete examination process. It connects students, faculty, examiners, examination staff and the Controller of Examinations through role-based pages and automated workflows.

Main Goals

- Reduce manual work and paperwork.
- Keep examination information in one place.
- Automate repetitive tasks such as allocation and result calculation.
- Provide clear approval and tracking workflows.
- Send important updates in real time.
- Provide useful reports and analytics.
- Protect confidential examination information.

4. Users

User	Main Activities
COE / Admin	Manage exams, approvals, timetable, halls, results, reports and overall monitoring.

User	Main Activities
Staff / Faculty	Manage student information, internal marks, attendance and assigned examination work.
Examiner / Valuator	Handle question paper/valuation tasks and enter marks.
Student	Register for exams, view timetable, hall ticket, results, certificates and revaluation.
System Admin	Manage users, roles, security and system settings.

5. Main Solution Modules

Module	Purpose
Dashboard	Shows the current examination status, important numbers, pending work and alerts.
Student Management	Stores and manages student academic and examination information.
Exam Management	Creates exams, manages registration and checks eligibility.
Timetable	Creates and publishes examination dates, times and sessions.
Centre & Hall Management	Manages examination centres, halls and capacities.
Seating Arrangement	Automatically assigns students to halls/seats and generates seating lists.
Staff & Examiner Management	Assigns invigilators and examiners to examination duties.
Question Paper Management	Manages question paper setting, approval and secure access.
Attendance	Records student attendance and generates absentee lists.
Valuation	Manages answer script allocation, valuation and mark entry.
Results	Calculates grades/GPA/CGPA and processes final results.
Revaluation	Allows eligible students to apply for revaluation or retotalling.
Certificates	Provides mark sheets, certificates and digital verification.
Notifications	Sends exam, timetable, hall, result and other important updates.
Reports	Generates examination, student, attendance, valuation and result reports.
Security	Controls access, protects data and records important activities.

6. Page Structure

The following pages form the main prototype. Each page has a specific purpose and should be simple for the user to understand.

1. Login Page

What the page contains	Why it is needed
- University/college logo and ExamSphere name - Username/email and password - Forgot password - OTP/MFA if required - Login button	Provides secure access and sends the user to the correct portal according to their role.

2. COE Dashboard

What the page contains	Why it is needed
• Total students registered • Upcoming examinations • Pending approvals • Exam readiness status • Upcoming timetable • Live notifications • Quick action buttons	Gives the COE a complete overview of what is happening and what needs attention.

3. Student Management Page

What the page contains	Why it is needed
• Search by register number/name • Student profile • Department and programme • Semester • Registered subjects • Eligibility status • Exam history	Allows examination staff to quickly view and manage student examination information.

4. Exam Registration Page

What the page contains	Why it is needed
• Exam name and semester • Eligible subjects • Registration status • Exam fee • Payment status • Submit/approve/reject options	Allows students to register and allows the COE to monitor and approve registrations.

5. Timetable Page

What the page contains	Why it is needed
• Exam date • Session/time • Subject • Programme/semester • Venue • Conflict warning • Publish button	Creates and publishes the official examination timetable.

6. Centre & Hall Page

What the page contains	Why it is needed
• Exam centres • Hall names • Hall capacity • Available seats • Allocated students • Centre readiness	Helps the COE manage examination locations and available capacity.

7. Seating Arrangement Page

What the page contains	Why it is needed
• Student list • Hall selection • Seat numbers • Register-number order/randomization • Hall layout • Generate PDF	Automatically creates a clear seating arrangement and reduces manual allocation.

8. Staff & Examiner Page

What the page contains	Why it is needed
• Staff/examiner list • Availability • Duty date • Assigned hall/subject • Workload • Duty confirmation	Assigns examination duties and helps balance staff workload.

9. Question Paper Page

What the page contains	Why it is needed
• Subject • Question paper status • Setter/moderator • Approval status • Secure document access • Release status	Manages confidential question papers with controlled access.

10. Attendance Page

What the page contains	Why it is needed
• Exam centre/hall • Student list • Present/absent • Late entry • Absentee list • Incident/malpractice option	Records examination attendance and provides real-time exam-day information.

11. Valuation Page

What the page contains	Why it is needed
• Valuation centre • Examiner • Answer script allocation • Marks entry • Verification • Submission/lock status	Tracks valuation and makes mark entry organized and auditable.

12. Result Processing Page

What the page contains	Why it is needed
• Student marks • Grades • GPA/CGPA • Pass/fail status • Arrear identification • Exception checking • Approval and publish	Automatically processes results and prepares them for final approval.

13. Student Results Page

What the page contains	Why it is needed
• Semester result • Subject-wise marks/grades • GPA/CGPA • Pass/fail status • Arrears • Download result	Provides students with a simple way to view their official results.

14. Revaluation Page

What the page contains	Why it is needed
• Eligible subjects • Application • Fee/payment • Request status • Updated marks • Final result	Allows students to request revaluation or retotalling and track the request.

15. Certificates Page

What the page contains	Why it is needed
• Available certificates • Request status • Download • Digital/QR verification • Certificate history	Provides students with secure access to examination documents.

16. Notifications Page

What the page contains	Why it is needed
• Announcements • Timetable updates • Hall changes • Exam reminders • Result notifications • Read/unread status	Keeps every user informed about important examination updates.

17. Reports Page

What the page contains	Why it is needed
• Exam registration report • Attendance report • Hall allocation report • Valuation report • Result report • Department performance • PDF/Excel download	Helps the COE generate useful reports for monitoring and decision-making.

7. Simple Examination Workflow

1. Student data is available in ExamSphere.
2. COE creates the examination.
3. Eligible students register and pay the examination fee.
4. COE verifies and approves registrations.
5. COE prepares and publishes the timetable.
6. Students are allocated to examination centres and halls.
7. Seating arrangement is generated automatically.
8. Invigilators and examiners receive their duties.
9. Exam-day attendance is recorded.
10. Answer scripts are sent for valuation.
11. Examiners enter and submit marks.
12. System calculates grades, GPA/CGPA and result status.
13. COE verifies and approves the results.

14. Results are published to students.
15. Students can apply for revaluation if eligible.
16. Certificates and mark sheets are provided digitally.

8. Expected Benefits

Benefit	How ExamSphere helps
Faster work	Automates registration, allocation, calculations and reports.
Less paperwork	Moves examination processes from paper to digital workflows.
Better accuracy	Reduces manual data-entry and calculation errors.
Transparency	Shows status, approvals and activities clearly.
Better communication	Sends real-time notifications to students and staff.
Better security	Provides role-based access and audit tracking.
Better decisions	Provides dashboards, reports and performance analytics.
Better student experience	Students can access most exam services from one portal.

9. Hackathon Demo Flow

For the final demonstration, show one complete journey rather than every module.

- COE logs in → Dashboard shows exam readiness.
- COE creates an exam → Students register.
- COE publishes timetable → Students receive notification.
- System allocates halls → Seating arrangement is generated.
- Invigilator records attendance.
- Examiner enters marks.
- System processes results.
- COE approves and publishes results.
- Student views result → applies for revaluation if needed.
- Student downloads certificate/mark sheet.

10. Final Prototype Idea

ExamSphere brings the complete Controller of Examination process into one secure and easy-to-use platform. The COE gets control and visibility, staff get organized workflows, examiners get focused tasks, and students get a simple self-service portal.

Core idea: Digitize → Automate → Monitor → Analyze → Notify → Improve